

Personal Statement

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I first became interested in mathematics when I taught myself calculus when I was twelve years old and decided that I wanted to pursue mathematics as my career. I participated in mathematics competitions throughout high school, where I first met W. Broughton at University of British Columbia Okanagan, who organized one of the competitions I participated. I decided to pursue maths at UBCO, where I was able to explore mathematics beyond the undergraduate curriculum and became particularly interested in topology. I pursued research opportunities in algebraic topology with P. Tsopmén  at UBCO, and quickly realized that this required me to learn substantial materials beyond my coursework. Therefore, I began studying category theory, homological algebra, and homotopy theory independently, through texts such as Rotman’s *Introduction to Homological Algebra* and Riehl’s *Categorical Homotopy Theory*. This self study introduced me to a style of mathematics that relies on the abstract formulation of concrete objects, and ultimately led to my first research project.

Under Tsopmén ’s supervision, I began working on a problem concerning the classification of homogeneous functors in manifold calculus. Manifold calculus, developed by Weiss and others, studies contravariant functors from the category of open subsets of a manifold to a homotopical target category. Weiss classifies the homogeneous functors of degree k value at homotopy type of a topological space A at the k disks with the homotopy classes of maps from the unordered configuration space of k points $F_k(M)$ into $B\mathrm{haut}(A)$, the classifying space of homotopy automorphisms of A . Tsopmén  and Stanley had subsequently constructed a simplicial set $\hat{A}_\bullet$ whose realization classifies homogeneous functors determined by a weak equivalent type of some object A in an arbitrary simplicial model category. Stanley and Tsopmén  conjectured that their construction was a generalization of the result of Weiss for arbitrary simplicial model categories, that is $B\mathrm{haut}(A) \simeq |\hat{A}_\bullet|$.

I became particularly interested in understanding the relationship between these two descriptions. The simplicial structure of $\hat{A}_\bullet$ contains complicated degeneracies that made a direct comparison with $B\mathrm{haut}(A)$ difficult. I proposed instead to discard the degeneracy maps and work with the underlying semisimplicial set. This was inspired by the work of C. Rourke and B. Sanderson in their works “ Δ -Sets”, which developed the comparison of semisimplicial sets with simplicial sets. During the initial few weeks of collaboration with Tsopmén , I exploited this approach by filling in several technical steps needed to establish the effectiveness of the semisimplicial approach and presenting the arguments to him. The semisimplicial formulation made the comparison substantially more transparent and ultimately led to the paper “On the Classifying Space of Homogeneous Functors”. This project was my first experience of developing a mathematical idea independently and then turning that idea into a rigorous argument.

My interest of mathematics extends to fields other than homotopy. During the summer of my third year, I completed a directed study supervised by E. Foxall at UBCO on mathematical aspects of quantum physics. We studied the spectral theorem for unbounded operators, Lie groups and Lie algebras and their representations, Hamiltonian mechanics on symplectic manifolds, and geometric quantization.

During working with Foxall, I became involved in his project on age-structured population model that studies a stochastic population process in which particles carry age labels and

reproduce and die according to age-dependent random variables. The project concerns the reproductive value, which describes the contribution of an individual at a given age to the future population, and in how this function changes under perturbations of the birth and death mechanisms. Since I had very little background in stochastic processes, I had to develop much of the necessary theory independently while working on the problem. This experience has strengthened my ability to perform research to an unfamiliar field.

My work in manifold calculus led to a broader interest in homotopy theory and, in particular, chromatic homotopy theory. I first encountered chromatic homotopy while studying formal group laws associated with complex-oriented cohomology theories through Rudyak's *On Thom Spectra, Orientability, and Cobordism*. The universal complex orientation of MU gives rise to the universal formal group law on MU_* , making the geometry of the moduli stack of formal groups $\mathcal{M}_{fg}$ closely connected with stable homotopy theory. This connection made me interested in the geometric interpretation of chromatic localization and in the relationship between the algebraic geometry of formal groups and the homotopy theory of spectra.

My development as a mathematician has been shaped not only by coursework and research, but also by opportunities to learn mathematics beyond the standard undergraduate curriculum. At UBC Okanagan, I explored advanced topics independently and through directed studies, eventually developing a particular interest in topology and pursuing research in algebraic topology. Evidences of my self-study include my notes on chromatic homotopy theory and solutions to exercises from R. Hartshorne's *Algebraic Geometry*, both available on my personal website. Independent research has also taught me to approach unfamiliar problems with patience and persistence and to communicate mathematical ideas clearly.

I have also valued opportunities to contribute to the mathematical community. I have participated in mathematics competitions since high school, later supporting high school competitors as a mentor, and participated in undergraduate mathematical activities such as the Putnam competition. I have presented my research at the UBCO Undergraduate Research Conference, attended seminars such as the eCHT research seminars, and summer schools where I engaged with researchers and students from different institutions. In particular, I participated as an undergraduate in the University of Regina's Summer School on Homotopy Colimits, where I discussed homotopy theory with graduate students and researchers.

These experiences have made me especially appreciative of collaborative mathematical communities and motivated me to contribute to them in graduate school.